

DEWDRIVE · REQUIREMENTS

DewDrive requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish (R11 target \$520; status from DWD-CAL-001 v0.3)
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); R7 restated for the drip screens; R8 and R10 wording; status from DWD-CAL-001 v0.2
0.3	2026-09-25	Amish Chadha	TRL 3 update (design point and budget decided by Amish in DWD-DDR-001; R11 redefined to \$500; status column from DWD-CAL-001)
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with a design point and concept status
0.1	2026-09-24	Amish Chadha	Initial scaffold

DewDrive requirements

These requirements were checked by calculation at TRL 3 in DWD-CAL-001 v0.3, for the design decided by Amish on 2026-09-25 (DWD-DDR-002): night air drawn down through the sealed trays, 25 wt % CaCl_2 , a drip screen under each tray and a fan cut-out above 70 % RH. The design meets eleven of thirteen. R11 (cost) is met since Amish approved the budget top-up to \$520 on 2026-09-26 (DWD-DDR-002, D15): the parts come to \$515. R6 and R10 can only be verified by test. The yield targets R1 and R2, missed by the v0.1 design, are now met. Targets are still proposals for review, not user-validated needs, and will be revised after co-design sessions.

■ Design point

Decided by Amish on 2026-09-25 (DWD-DDR-001, D8). Unless a requirement says otherwise, it applies at this design point:

- Night: 10 h adsorption at 20 °C air and 40 % RH (6.92 g/m³ of vapour).
- Day: 6.0 kWh/m² of sun on the tilted aperture, of which 4.24 kWh/m² falls in the 09:00 to 15:00 window (DWD-CAL-001, C1); air at 35 °C.
- Low-humidity case: night air at 20 °C and 25 % RH, same sun.

■ Requirements

Table 1. Requirements and TRL 3 status. Values and tags are from DWD-CAL-001 v0.3, Table 4.

ID	Requirement	Target	Verification	TRL 3 status
R1	Produce water at the design point	0.5 L per day or more collected, per unit	Cyclic mass and heat balance (DWD-CAL-001); later field test	Met: 0.57 L per day (C3); v0.1 design 0.20 L
R2	Produce water in dry nights	0.25 L per day or more at 25 % night RH	Same	Met: 0.45 L per day (C4); v0.1 design 0.09 L
R3	Use only the sun for regeneration	No fuel or grid power; bed reaches 75 °C or more by midday on a clear day at 35 °C ambient	Collector heat balance	Met: bed about 112 °C at noon (C2)
R4	Run its own electrical loads	Fan and logger need 30 Wh per day or less, supplied by the on-board PV panel, with 2 days or more of battery autonomy	Energy budget	Met: 23.6 Wh per day against 35 Wh per day; 2.6 days of autonomy (F1, F2)
R5	Keep the condenser cool	Condenser 15 K or less above ambient at the peak of desorption	Condenser heat balance	Met: 12.1 K (C2)
R6	Keep the water clean	All wetted parts food-grade; no salt carry-over (chloride in the collected water below 250 mg/L)	Materials review; later laboratory water test	Not verifiable at TRL 3. Materials chosen to meet it
R7	Contain the salt	No liquid CaCl ₂ solution leaves the tray and drip-screen assembly or reaches the condenser, even after a night at 90 % RH (restated by DWD-DDR-002, D11; was "leaves the trays")	Pore volume against solution volume; sump volume; later chamber test	Met on paper: the pores fill only above 71 % RH (E2); one 90 % RH night fills 43 % of them (E3), three in a row 50 % with the fan rule and 63 % without it (E4, E5); the sumps hold 0.58 L as a backstop (E6)
R8	Be simple to operate	Two user actions per day (open flaps in the evening, close them in the morning), 5 min per day or less	Task analysis; co-design sessions	Met by design; the fan timer and humidity cut-out need no user action
R9	Be portable and quick to set up	Box 35 kg or less (two-person lift); stand separable; set up by two people in 30 min or less	Mass from model volumes; later trial	Met: box 33.4 kg, 24.0 kg with trays, sorbent and screens out; total 53.9 kg (G1, G2). Set-up time needs a trial
R10	Survive the site	Stable in 20 m/s (72 km/h) wind with two ground anchors (default, DWD-DDR-002, D13) or ballast; UV-stable glazing; sorbent 300 cycles or more with 20 % or less loss of capacity	Wind load calculation; supplier data; later cycling test	Not verifiable at TRL 3. Wind: two anchors of 114 N each, or 46 kg of ballast (H2 to H4)
R11	Cost	Parts \$520 or less (<code>project.yaml</code> budget, raised from \$400 to \$500 by Amish on 2026-09-25, DWD-DDR-001 D1; budget top-up to \$520 approved by Amish on 2026-09-26, DWD-DDR-002 D15)	Priced BOM	Met: \$515 (J1), \$5 under
R12	Record performance	Log air temperature and RH, bed and condenser temperature every 5 min for	Storage estimate; design review	Met: 8,640 records, 0.41 MB (I1)

ID	Requirement	Target	Verification	TRL 3 status
		30 days or more; daily water mass recorded by weighing		
R13	Protect users	External surfaces a person would touch in normal use stay at 60 °C or less; the glazing is labelled hot; no exposed conductor above 60 V DC	Surface temperature estimate; design review	Met: glazing outer skin about 57 °C at stagnation, fins up to 47 °C (D2, D3); 12.8 V DC

■ Assumptions

- The composite (25 wt % CaCl₂: 1.0 kg in 3.0 kg of mesoporous silica gel) is modelled from bulk salt thermodynamics: at equilibrium it holds 0.397 g/g at 40 % RH and 0.258 g/g at 25 % RH (DWD-CAL-001, A4). The yield is set by how fast the bed takes up and gives off water, not by these capacities. Measured isotherms and uptake rates of the actual composite must replace the model.
- The day ends with the salt dried to about the dihydrate; the night starts with the bed at air temperature.
- With the fan stopped above 70 % RH, natural airflow through the open flaps is taken as 4 m³/h (10 % of the fan flow); this is an assumption to check by test.
- One unit is a supplement: about 0.57 L per day is about one fifth of one person's survival drinking need of 2.5 to 3 L per day (WHO technical note 9).

Safety: Requirements R6, R7, R10 and R13 exist because of hazards: untreated water, a corrosive salt solution, wind loads on a 1.1 m² panel, and hot surfaces. They cannot be waived without a documented decision by Amish.